

WEEK 1 | LAB REPORT

SIEM Architecture & Log Onboarding

SIEM Detection Engineering, Alert Triage & SOC Operations

PLATFORM

Wazuh 4.14.9 (all-in-one) on Ubuntu Server 26.04 LTS

LAB DATE

22 Oct 2026

3	1	223	90 d
CORE SERVICES RUNNING	AGENT ONBOARDED	EVENTS IN 24 H	ALERT RETENTION

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AT A GLANCE

3	1	223	90 d
CORE SERVICES RUNNING	AGENT ONBOARDED	EVENTS IN 24 H	ALERT RETENTION

Scope

Platform: Wazuh 4.14.9 (all-in-one) on Ubuntu Server 26.04 LTS. Lab date: 22 Oct 2026.

This report covers the SIEM architecture (lab vs. a medium-organization design with deployment verification), the data source inventory and field mapping, the retention plan (measured data, design and ISM policy), and an evidence index of the screenshots.

SECTION 01

1. SIEM Architecture

The SIEM is built on Wazuh, which combines the Manager (decoders, rules, MITRE and compliance mapping, active response), the Indexer (OpenSearch-based storage and search) and the Dashboard (analyst UI). Agents on endpoints and other sources ship events to the Manager, which analyses them and forwards alerts to the Indexer through Filebeat.

Manager
Decoders, rules, MITRE and compliance mapping, active response.

Indexer
OpenSearch-based storage and search.

Dashboard
Analyst UI.

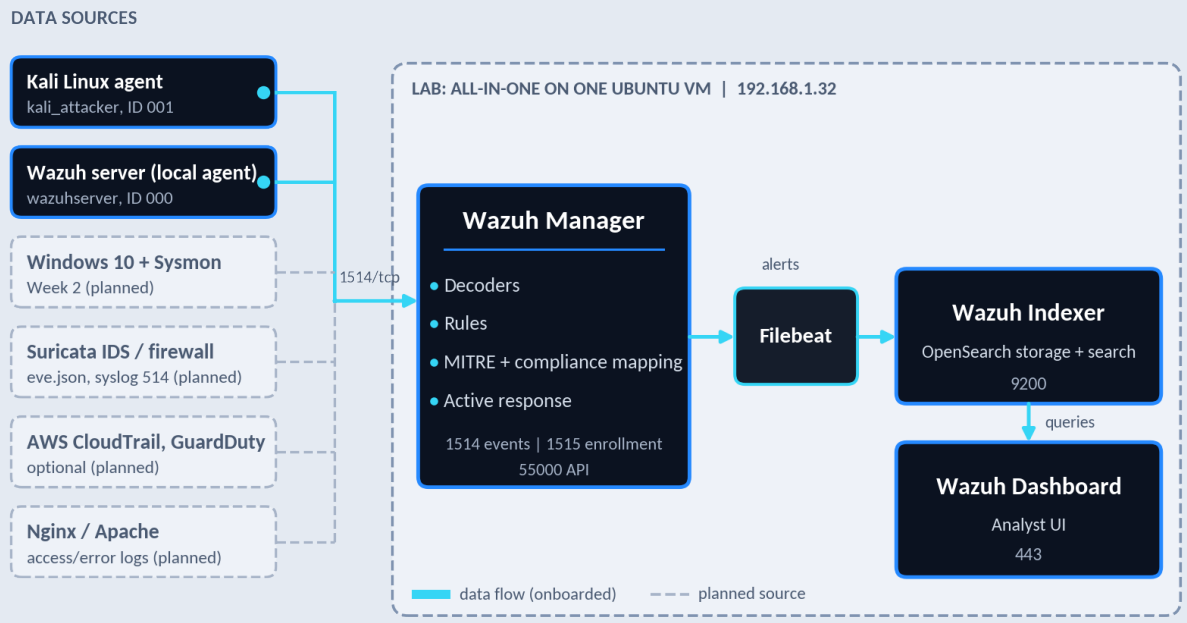


Diagram A *Component view of the lab deployment and planned sources.*

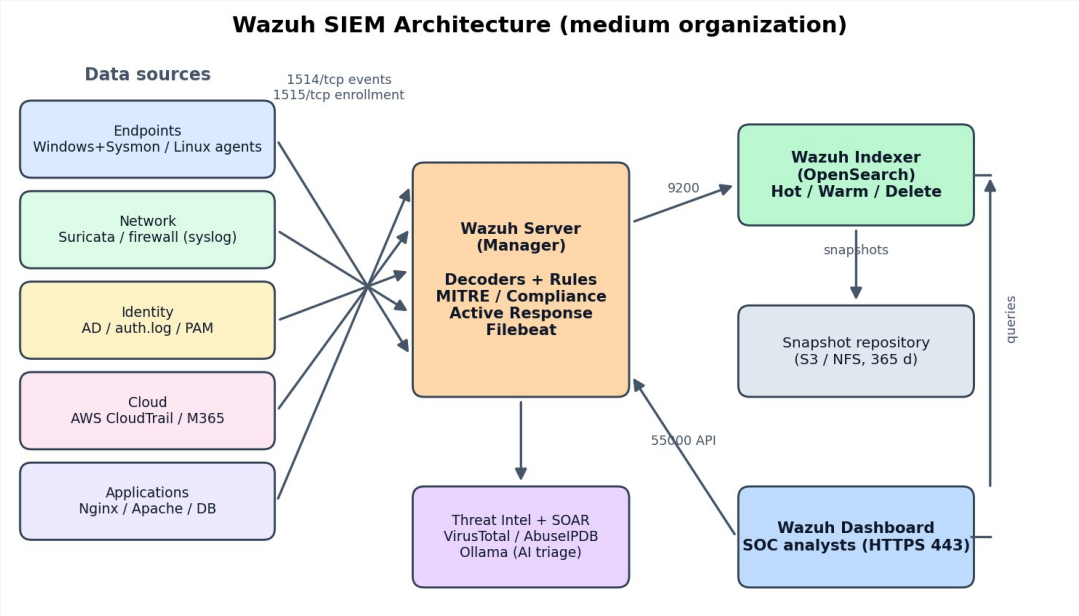


Figure 1 *Wazuh SIEM architecture*

WHAT THIS SCREEN DOES

Overview of how the SIEM is wired. Endpoint agents send events to the Wazuh Manager, which decodes them, applies detection rules and enriches them with MITRE ATT&CK and compliance tags. Alerts are then forwarded by Filebeat to the Indexer for storage and searched by analysts in the Dashboard.

1.1 Lab implementation vs. medium-organization design

ITEM	LAB (IMPLEMENTED)	MEDIUM ORGANIZATION (DESIGN)
Deployment	All-in-one on one Ubuntu VM (VirtualBox, bridged network, 192.168.1.32)	Distributed: Manager cluster (2 to 3 nodes), Indexer cluster (3 nodes), 1 to 2 Dashboards behind a load balancer
Resources	5 vCPU, 5.3 GiB RAM, 48 GB disk (below the 8 GB RAM recommendation; installer run with -i)	Indexer nodes: 8 vCPU, 32 GB RAM, SSD; Manager nodes: 8 vCPU, 16 GB RAM
Sources	1 agent (Kali Linux) plus the server itself	About 500 endpoints plus network, identity, cloud and application sources
Replicas	0 (single node, all indices Green)	1 replica per index for high availability
Ports	1514/tcp events, 1515/tcp enrollment, 55000 API, 9200 Indexer, 443 Dashboard	Same, restricted by firewall and segmented by VLAN

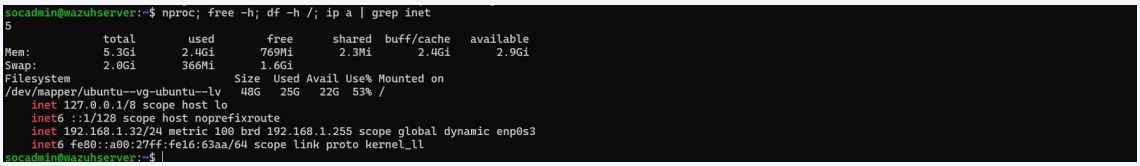


Figure 2 *W1-02 – Server specs: nproc, free -h, df -h, IP 192.168.1.32*

WHAT THIS SCREEN DOES

Checks the server resources with one command line: nproc (CPU cores), free -h (RAM), df -h / (disk) and ip a (network). It proves the Wazuh VM has 5 vCPUs, 5.3 GiB RAM, a 48 GB disk (53% used) and the static-looking address 192.168.1.32 on a bridged interface, which is the address agents will report to.

1.2 Deployment verification

- Indexer, Manager and Dashboard services are all active (running) (Figure 4, W1-05).
- Agent kali_attacker (ID 001, 192.168.1.187, v4.14.9) is Active in Endpoints Summary (Figure 7, W1-08).
- 223 events from the agent were visible in Threat Hunting within 24 hours (Figure 10, W1-09).
- The installer printed a warning that Ubuntu 26.04 is not in the officially supported list. The installation completed and all services run normally (Figure 3, W1-04).

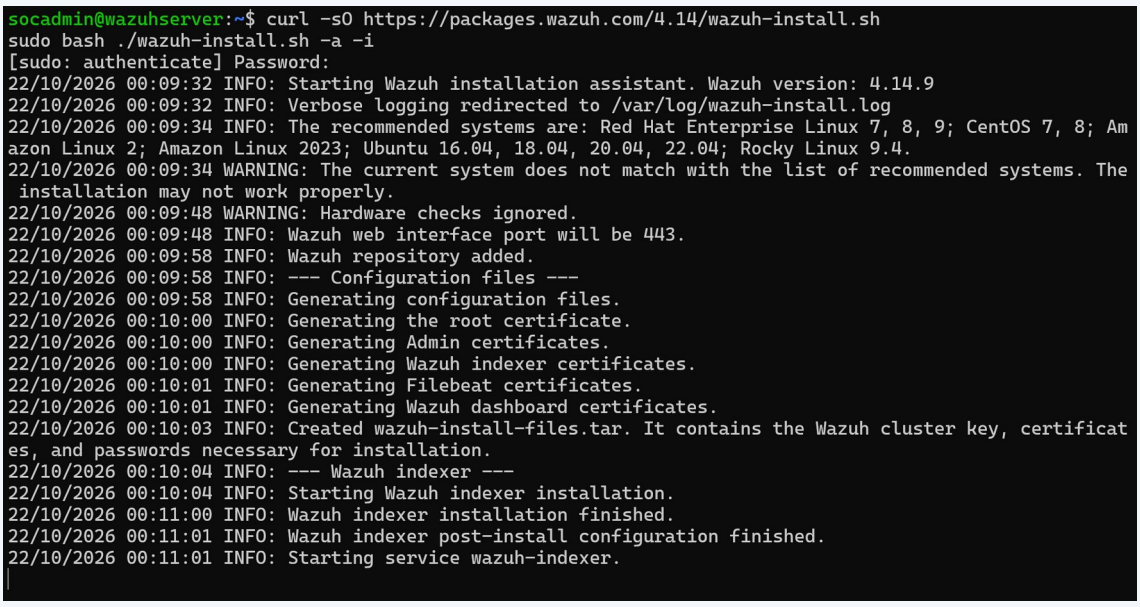


Figure 3 *W1-04 – Wazuh installer output (version 4.14.9, warnings)*

WHAT THIS SCREEN DOES

The Wazuh all-in-one installer running (wazuh-install.sh -a -i). It prints a warning that Ubuntu 26.04 is not an officially supported OS and that hardware checks were skipped (-i flag), then generates the root, admin, indexer, Filebeat and dashboard certificates and starts installing the Indexer. Dashboard port is 443.

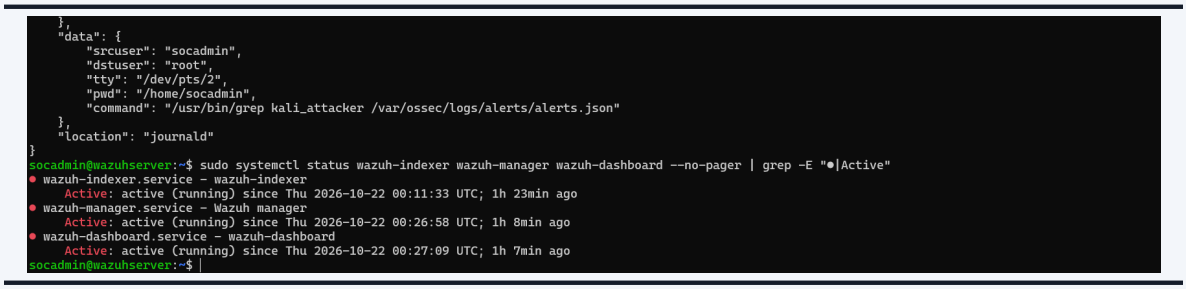


Figure 4 *W1-05 – Indexer, Manager and Dashboard services active (running)*

WHAT THIS SCREEN DOES

Verifies the deployment with systemctl status. All three core services, wazuh-indexer, wazuh-manager and wazuh-dashboard, show active (running). The JSON fragment above the command is a real sudo alert (srcuser socadmin, dstuser root) produced by the server's own logs.

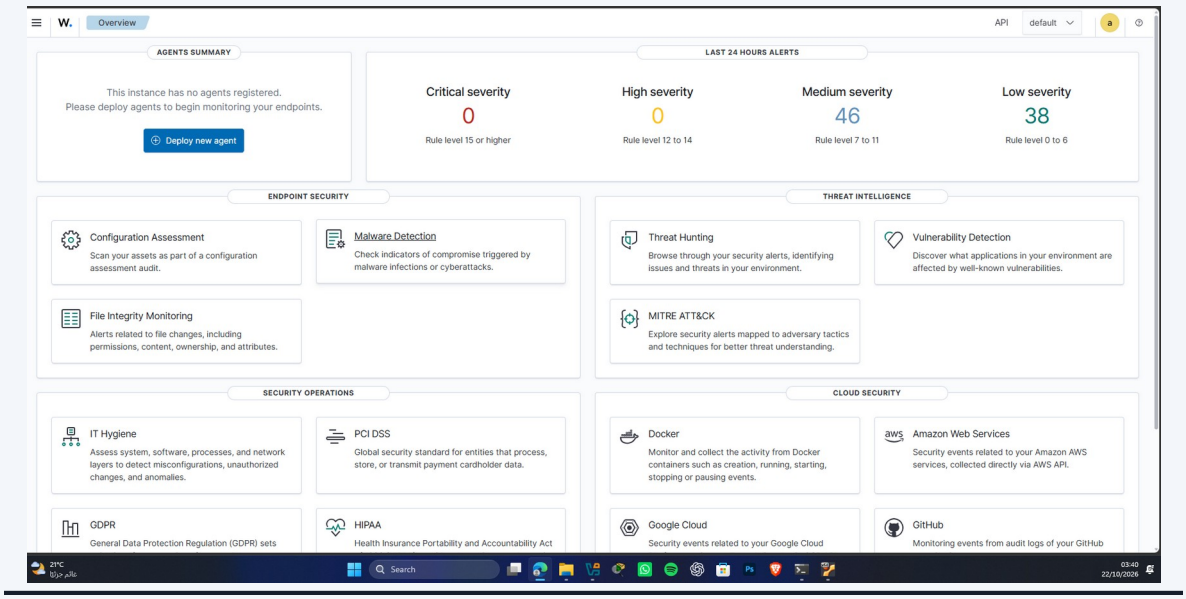


Figure 5 *W1-06 – Dashboard home before any agent was registered*

WHAT THIS SCREEN DOES

The Dashboard home right after installation and before any agent was added. The Agents Summary says no agents are registered, while the last-24-hours alert counters (Medium 46, Low 38) come from the Wazuh server monitoring itself. It also shows the available modules such as Threat Hunting, MITRE ATT&CK and Vulnerability Detection.

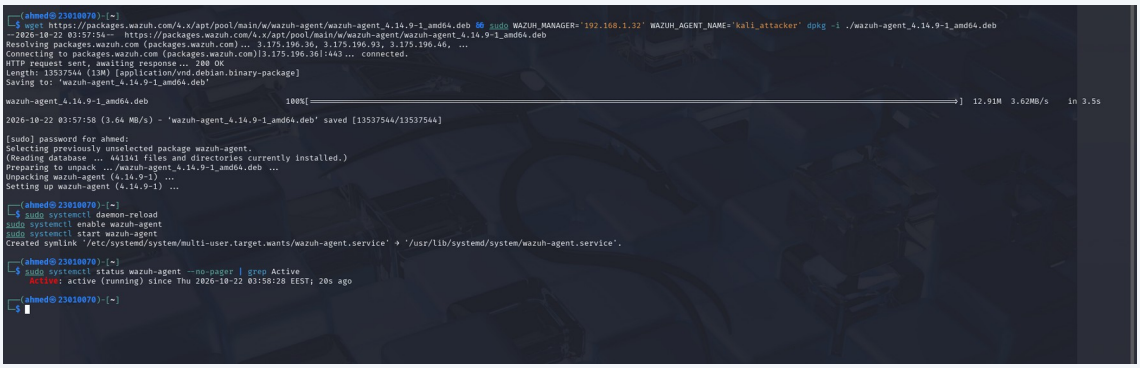


Figure 6 *W1-07b – Agent installation and service start on Kali*

WHAT THIS SCREEN DOES

Onboarding the Kali machine. The agent .deb package is downloaded and installed with dpkg while WAZUH_MANAGER points to 192.168.1.32 and WAZUH_AGENT_NAME sets the name kali_attacker. Then systemctl reloads, enables and starts wazuh-agent, and the status confirms it is active (running).

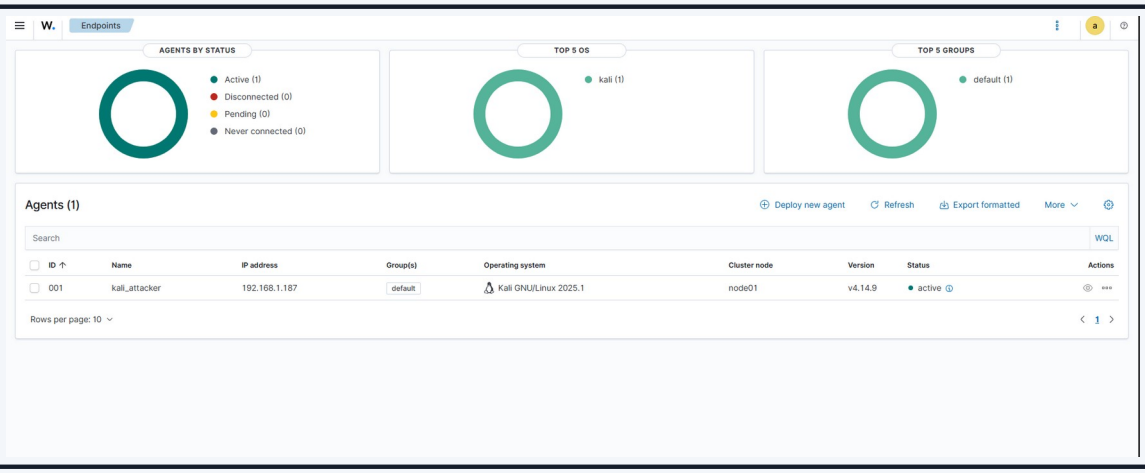


Figure 7 *W1-08 – kali_attacker shown as Active in Endpoints Summary*

WHAT THIS SCREEN DOES

Endpoints Summary after enrollment. One agent is registered: ID 001, kali_attacker, 192.168.1.187, group default, Kali GNU/Linux 2025.1, version 4.14.9, status active. The donut charts show 1 active, 0 disconnected, 0 pending and 0 never connected.

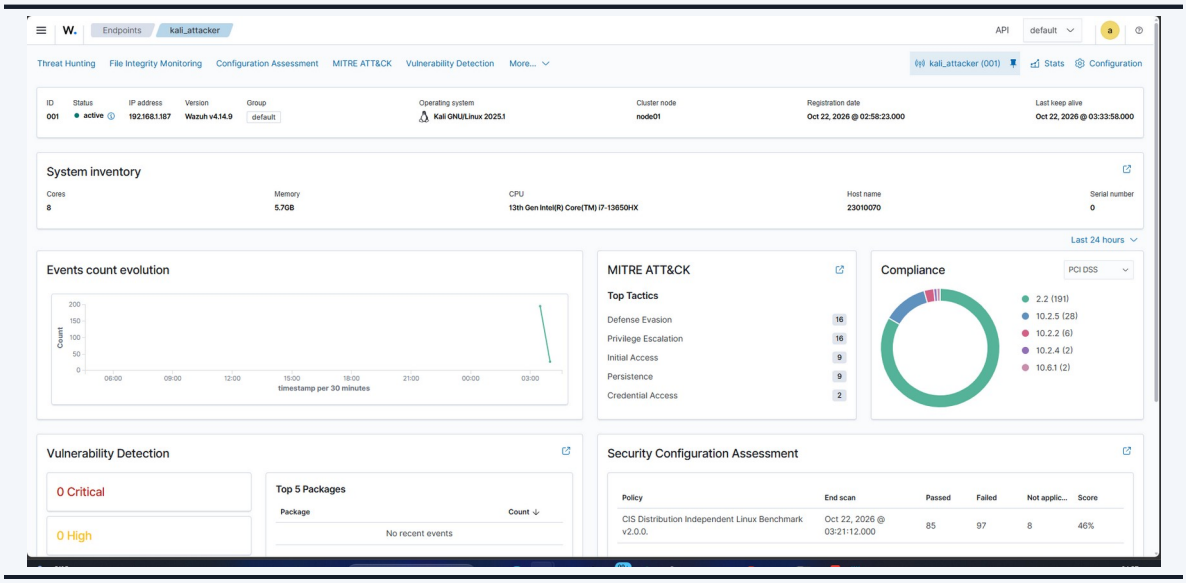


Figure 8 *W1-08b – Agent overview: inventory, MITRE ATT&CK, compliance, SCA*

WHAT THIS SCREEN DOES

Agent overview page for kali_attacker. It shows inventory (8 cores, 5.7 GB RAM, Core i7-13650HX), the top MITRE ATT&CK tactics seen (Defense Evasion 16, Privilege Escalation 16), PCI DSS compliance counters, 0 critical/high vulnerabilities, and the first CIS benchmark scan: 85 checks passed, 97 failed, score 46%.

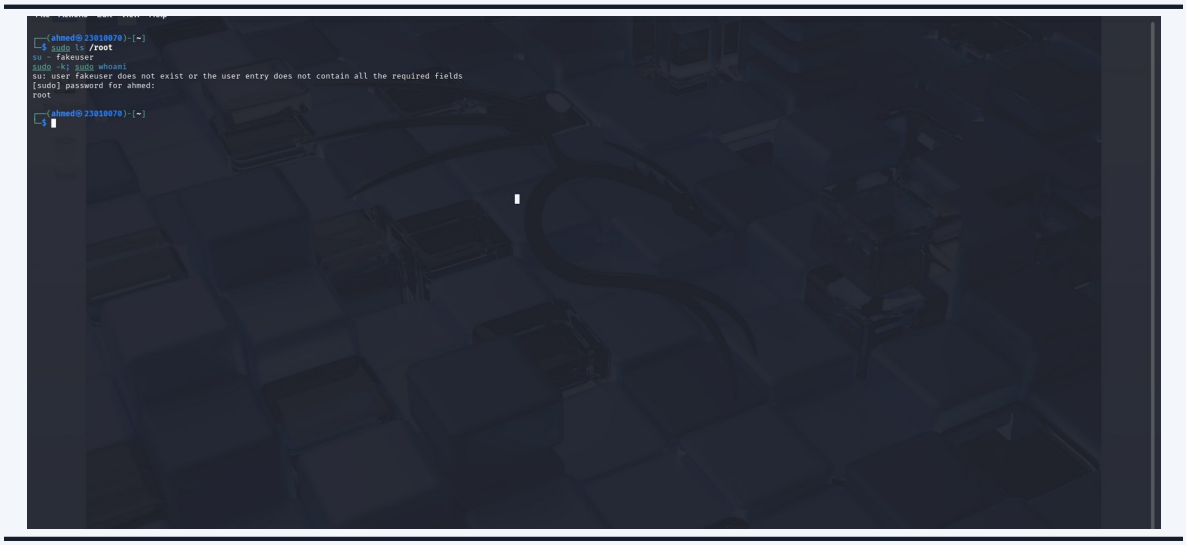


Figure 9 *W1-09a – Test events generated on Kali (su, sudo)*

WHAT THIS SCREEN DOES

Test activity generated on Kali to create log events. It runs sudo ls /root, a su to a non-existent account (fakeuser, which fails) and sudo -k followed by sudo whoami (password asked, returns root). These produce PAM session and sudo events that Wazuh should collect.

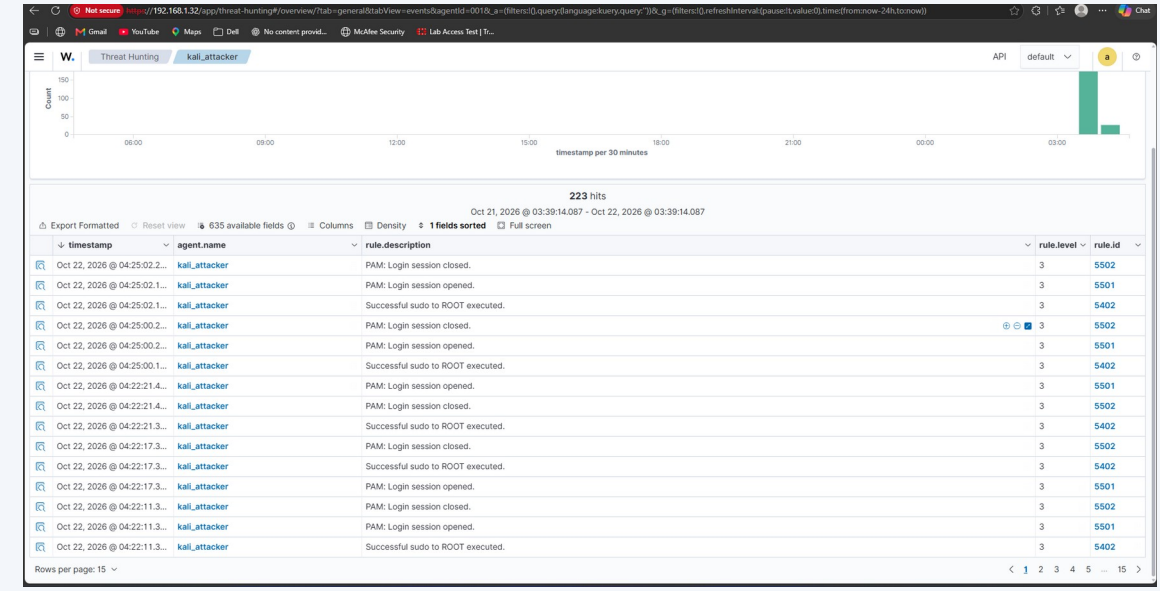


Figure 10 *W1-09 – 223 events from the agent in Threat Hunting*

WHAT THIS SCREEN DOES

Threat Hunting view for agent 001 over the last 24 hours: 223 hits. The table lists the events from the test, such as rule 5501 (PAM login session opened), 5502 (session closed) and 5402 (successful sudo to ROOT), all level 3. It proves logs travel from the agent to the Manager and Indexer.

SECTION 02

2. Data Source Inventory

Status: **Onboarded** = sending logs now; **Planned** = designed for the next phase and needed by Week 2 detection use cases.

CATEGORY	SOURCE	COLLECTION METHOD	KEY LOG TYPES / EVENTS	STATUS
Endpoints	Kali Linux (kali_attacker)	Wazuh agent (DEB)	journald/auth: sudo, su, PAM sessions; SCA (CIS benchmark); syscollector inventory; FIM	**Onboarded**
Endpoints	Wazuh server (wazuhserver, ID 000)	Local agent	sudo, PAM, manager/indexer service logs, vulnerability and inventory states	**Onboarded**
Endpoints	Windows 10 VM + Sysmon	Wazuh agent (MSI) + Sysmon channel	Security 4624/4625/4672/4688, Sysmon 1/3/11/13, PowerShell 4104	**Planned (Week 2)**
Network	Suricata IDS / firewall	Wazuh agent reading eve.json; syslog 514	Alerts, flows, DNS, HTTP, blocked connections	**Planned**
Identity	Linux auth (sshd, PAM) / Active Directory	Agent journald; Windows Security events	Logon success/failure, privilege use, account changes	**Partially onboarded (PAM/sudo)**
Cloud	AWS CloudTrail, GuardDuty	Wazuh AWS module (S3 + SQS)	ConsoleLogin, IAM changes, API calls	**Planned (optional)**
Applications	Nginx / Apache web server	Agent logcollector (access/error logs)	HTTP status, source IP, URI, user agent	**Planned**

2.1 Field mapping (raw log to normalized fields)

Real event from the lab (Wazuh server, rule 5402, Successful sudo to ROOT). Pipeline: raw log, predecoder, decoder, rules, alert JSON, Indexer, Dashboard (screenshots W1-10 and W1-10b).

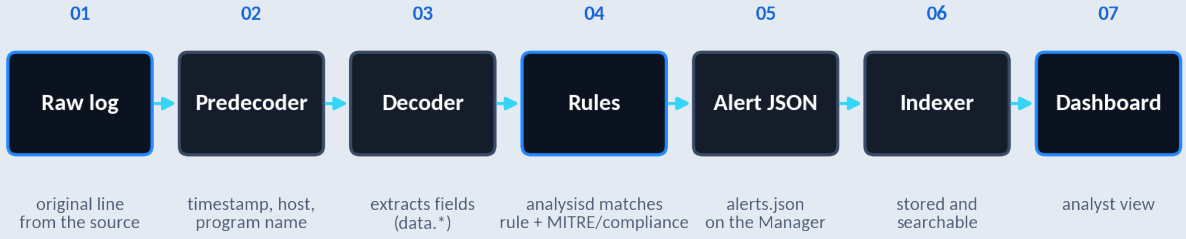


Diagram B *Processing pipeline from raw log to analyst view.*

```
Oct 22 01:29:57 wazuhserver sudo[59767]: socadmin : TTY=/dev/pts/2 ;
PWD=/home/socadmin ;
USER=root ; COMMAND=/usr/bin/grep kali_attacker /var/ossec/logs/alerts/alerts.json
```

RAW LOG PART	STAGE	NORMALIZED FIELD	VALUE
Oct 22 01:29:57	predecoder	predecoder.timestamp	Oct 22 01:29:57
wazuhserver	predecoder	predecoder.hostname	wazuhserver
sudo[59767]	predecoder / decoder	predecoder.program_name, decoder.name	sudo
socadmin	decoder	data.srcuser	socadmin
USER=root	decoder	data.dstuser	root
TTY=/dev/pts/2	decoder	data.tty	/dev/pts/2
PWD=/home/socadmin	decoder	data.pwd	/home/socadmin
COMMAND=...	decoder	data.command	/usr/bin/grep kali_attacker ...
Source of log	agent	location	journald
Sending agent	manager	agent.id / agent.name	000 / wazuhserver
Matched rule	analysisd	rule.id / rule.level / rule.description	5402 / 3 / Successful sudo to ROOT executed.
Rule metadata	analysisd	rule.mitre.id / tactic / technique	T1548.003 / Privilege Escalation, Defense Evasion / Sudo and Sudo Caching
Rule metadata	analysisd	rule.pci_dss, nist_800_53, gdpr, hipaa, tsc	10.2.5, 10.2.2 / AU.14, AC.7, AC.6 / ...

Kali example (rule 5502, PAM: Login session closed), same pipeline:

```
Oct 22 01:25:01 23010070 sudo[7031]: pam_unix(sudo:session): session closed for user
root
-> agent.name=kali_attacker, agent.ip=192.168.1.187, predecoder.hostname=23010070,
rule.id=5502, rule.level=3
```

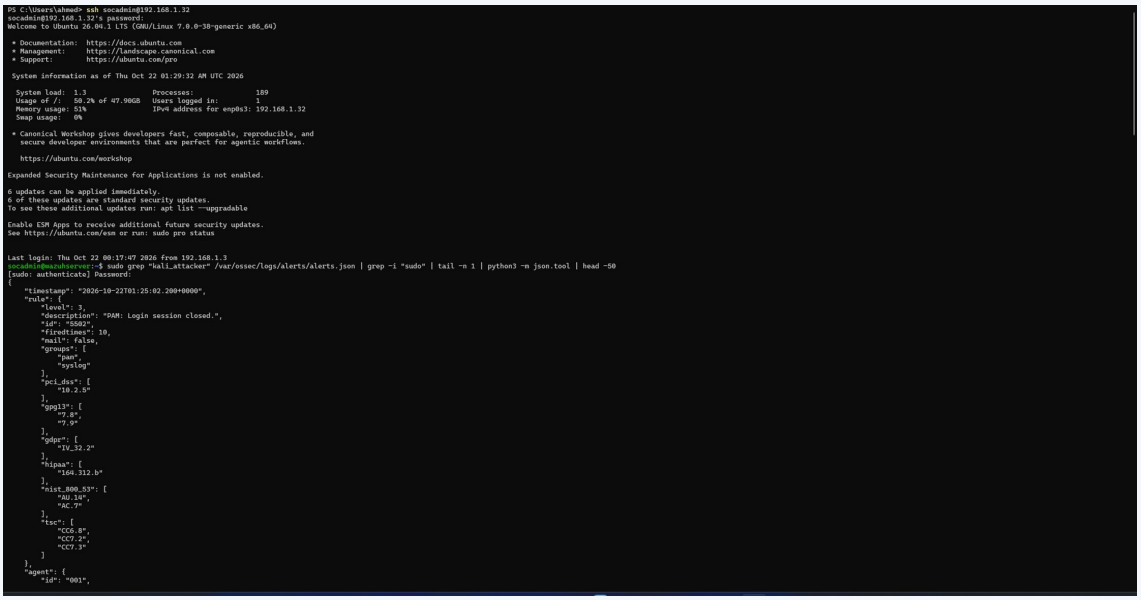


Figure 11 *W1-10 – Kali alert JSON (rule 5502, PAM: Login session closed)*

WHAT THIS SCREEN DOES

Raw alert from the Manager's alerts.json, pretty-printed from an SSH session to the server (filtered for kali_attacker and sudo). It shows rule 5502 (PAM: Login session closed), level 3, fired 10 times, with its groups and compliance mappings (PCI DSS 10.2.5, GPG13, GDPR, HIPAA, NIST 800-53, TSC) and the agent block (001, kali_attacker).

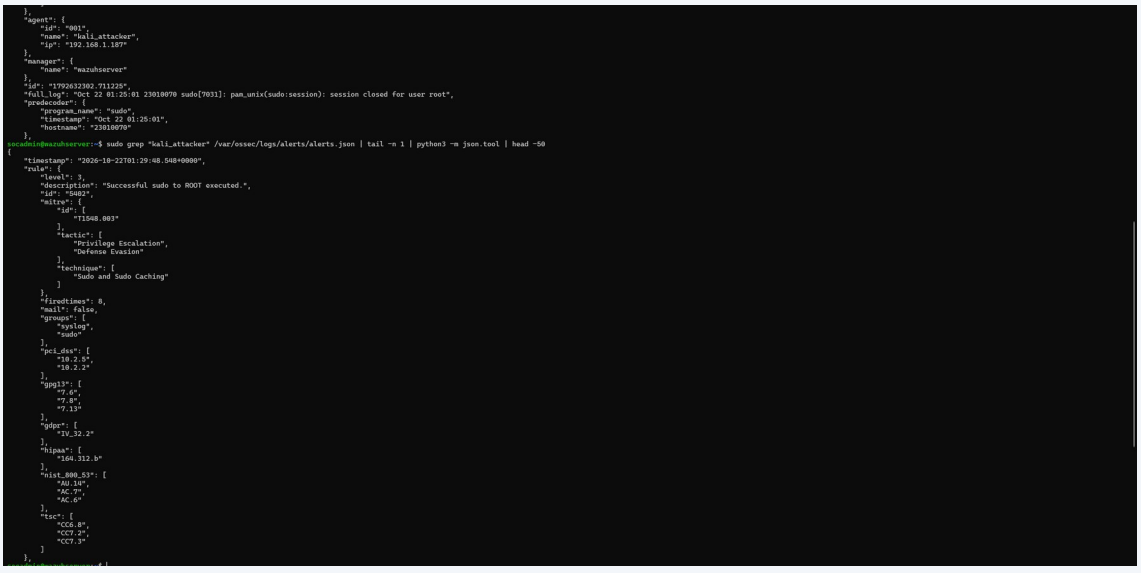


Figure 12 *W1-10b – Alert JSON with normalized fields (rule 5402, MITRE T1548.003)*

WHAT THIS SCREEN DOES

The rest of the Kali alert and the newest alert in the file. It shows the manager, the full_log line and the predecoder fields (program sudo, hostname 23010070). Below it, rule 5402 (Successful sudo to ROOT) adds MITRE data: T1548.003, tactics Privilege Escalation and Defense Evasion, technique Sudo and Sudo Caching.

SECTION 03

3. Retention Plan

3.1 Measured data from the lab

385	947.7 KB	~2.5 KB	17
ALERTS IN ONE DAY	DAILY ALERT INDEX SIZE	AVERAGE ALERT SIZE	INDICES, ALL GREEN

MEASUREMENT	VALUE
Alerts index	wazuh-alerts-4.x-2026.10.22 (3 primary shards, 0 replicas)
Size / documents (one day, one agent)	947.7 KB / 385 alerts
Average alert size	about 2.5 KB
Total indices at measurement time	17, all Green
Largest non-alert index	wazuh-states-inventory-packages (961 KB, 2,886 docs)
Lifecycle policy before this work	None (Managed by policy = No for all indices)

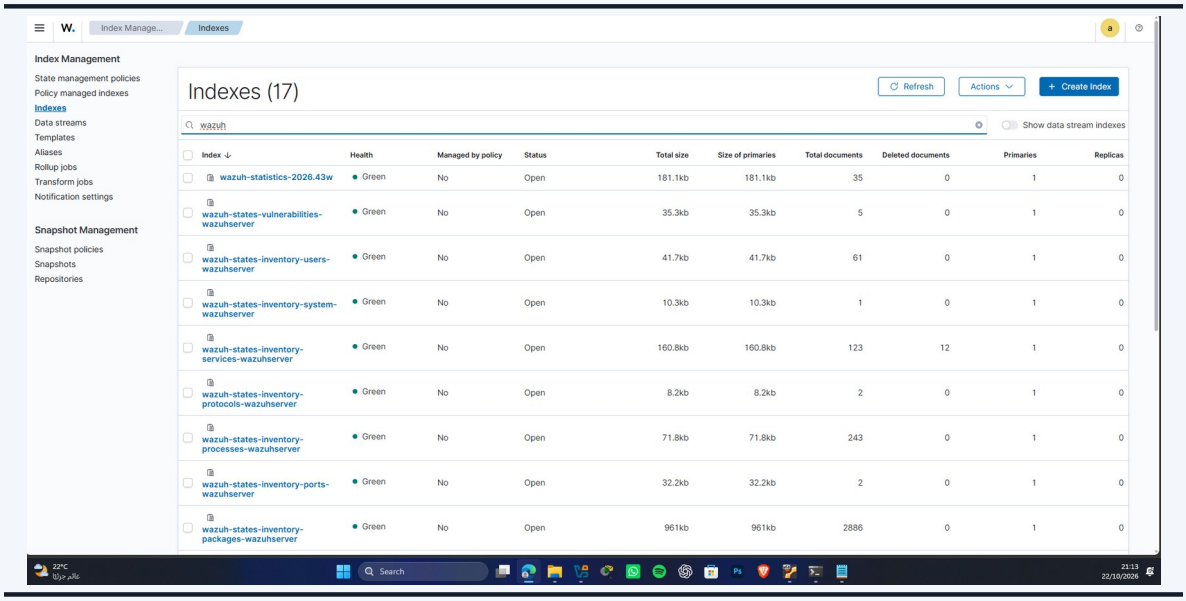


Figure 13 *W1-12 – Indices list and sizes*

WHAT THIS SCREEN DOES

Index Management listing 17 wazuh indices, all Green and all with Managed by policy = No at this point. It gives the measured sizes used in the plan, for example wazuh-states-inventory-packages at 961 KB with 2,886 documents, and shows there was no lifecycle policy yet.

3.2 Design for a medium organization (planning estimates)

> *Assumptions (to be replaced with measured values after onboarding more sources): 500 endpoints, about 2 GB/day of raw logs, alerts about 150 MB/day, one replica in production.*

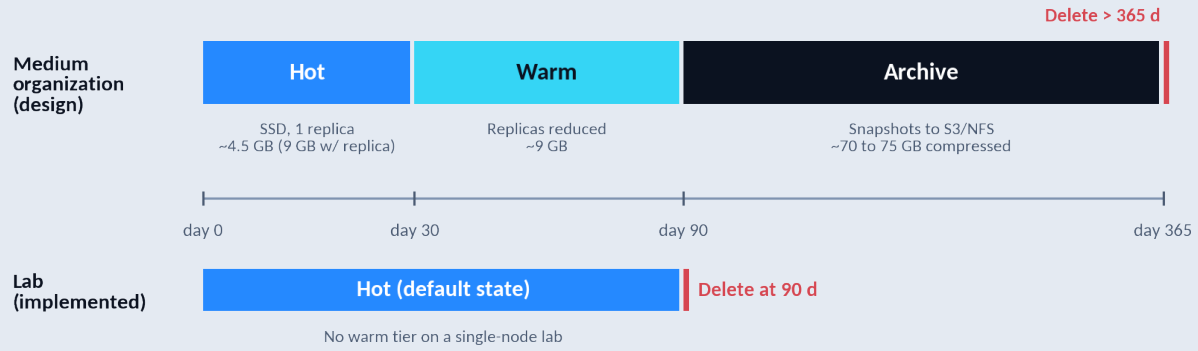


Diagram C *Retention tiers, medium-organization design vs. lab policy (time axis not to scale).*

TIER	AGE	CONTENT / PURPOSE	STORAGE	ESTIMATED SIZE
Hot	0 to 30 days	Fully searchable alerts for triage and live detection	SSD, 1 replica	about 4.5 GB (9 GB with replica)
Warm	31 to 90 days	Investigations, threat hunting, KPI trends	Cheaper disk, replicas reduced	about 9 GB
Archive	91 to 365 days	Compliance evidence (PCI DSS, ISO 27001), restore on demand	Snapshots to S3/NFS, compressed	about 70 to 75 GB (compressed raw logs)
Delete	over 365 days	Removed unless legal hold applies	-	-

3.3 Policy implemented in the lab

The lab is single-node with no warm tier, so a simplified Index State Management (ISM) policy was created and applied: alerts stay in the hot state and are deleted at 90 days. The policy auto-attaches to new wazuh-alerts-* indices through ism_template, and was applied manually to the existing index (Managed by policy changed from No to Yes, screenshots W1-12c1, W1-12c2, W1-12d).

```
{
  "policy": {
    "description": "SOC lab retention: hot 30d, delete after 90d",
    "default_state": "hot",
    "states": [
      { "name": "hot", "actions": [],
        "transitions": [ { "state_name": "delete",
          "conditions": { "min_index_age": "90d" } } ] },
      { "name": "delete", "actions": [ { "delete": {} } ], "transitions": [] }
    ],
    "ism_template": [ { "index_patterns": ["wazuh-alerts-*"], "priority": 100 } ]
  }
}
```

PRODUCTION EXTENSION

Add a warm state at 30 days (reduce replicas, force merge), a snapshot policy to an S3/NFS repository at 90 days, and delete at 365 days. Apply the same pattern to wazuh-archives-* if archiving is enabled.

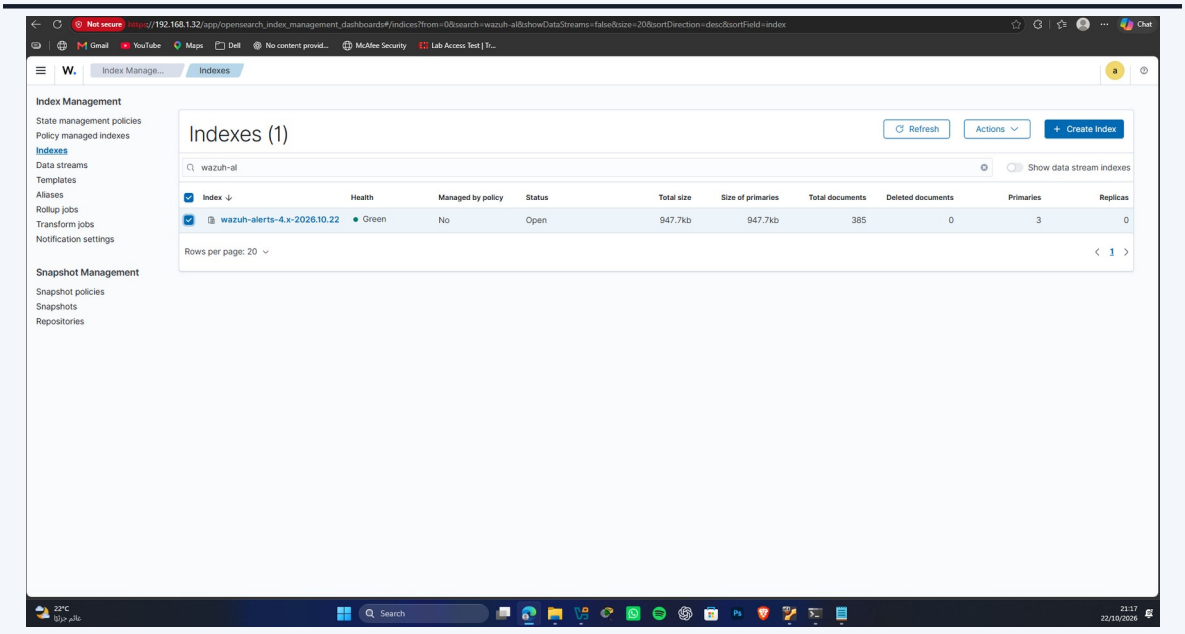


Figure 14 *W1-12c1 – Alerts index before policy (Managed by policy = No)*

WHAT THIS SCREEN DOES

Filtered to the alerts index wazuh-alerts-4.x-2026.10.22: 947.7 KB, 385 documents, 3 primary shards, 0 replicas, Managed by policy = No. This is the baseline used for the average alert size (about 2.5 KB) and the state before retention was applied.

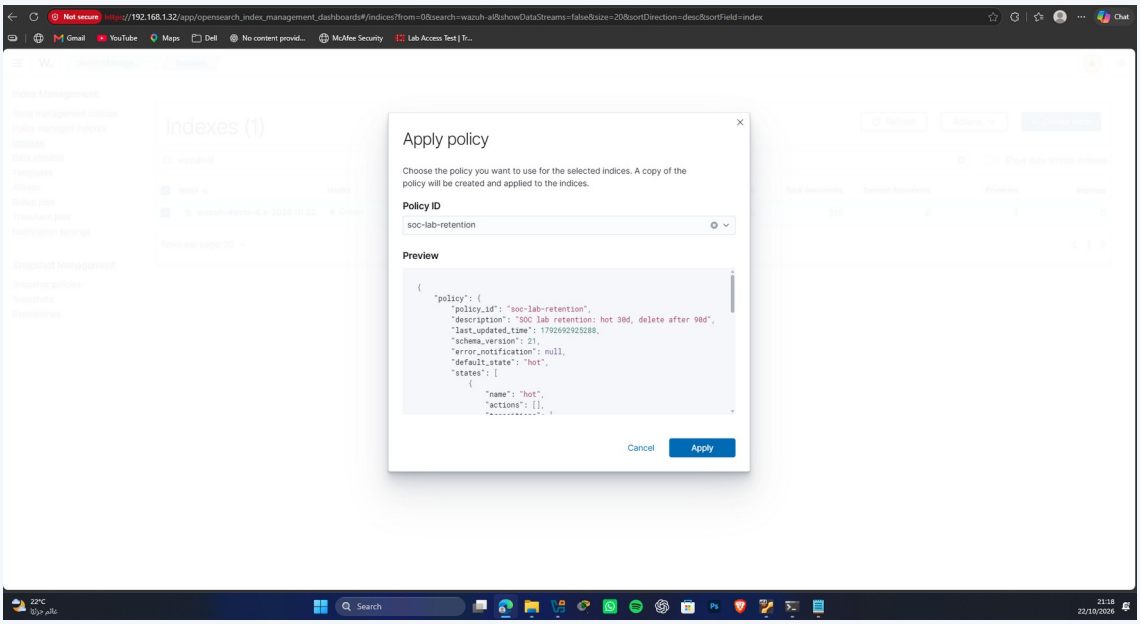


Figure 15 *W1-12c2 – Apply policy dialog (soc-lab-retention)*

WHAT THIS SCREEN DOES

The Apply policy dialog. Policy ID soc-lab-retention is selected and the preview shows its definition (description: hot 30d, delete after 90d, default state hot). Clicking Apply attaches a copy of the policy to the selected index.

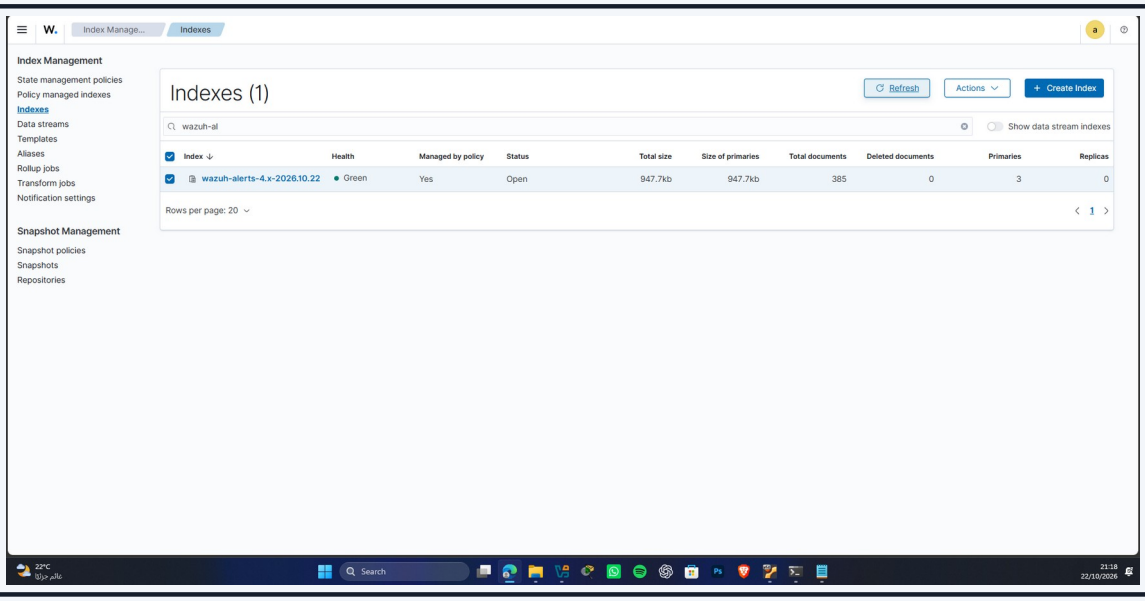


Figure 16 *W1-12d – Policy applied (Managed by policy = Yes)*

WHAT THIS SCREEN DOES

Same index after applying the policy. Managed by policy now shows Yes, confirming the retention policy is active on wazuh-alerts-4.x-2026.10.22 (the ism_template also attaches it to future daily alert indices automatically).

SECTION 04

4. Evidence Index

FIGURE	FILE	SHOWS
2	W1-02-server-specs.png	nproc, free -h, df -h, server IP 192.168.1.32
3	W1-04-installer-output.png	Wazuh installer progress (version 4.14.9, warnings)
4	W1-05-services-running.png	Indexer, Manager and Dashboard active (running)
5	W1-06-dashboard-home-before-agent.png	Dashboard before any agent was registered
6	W1-07b-kali-agent-install.png	Agent installation and service start on Kali
7	W1-08-endpoints-active.png	kali_attacker Active in Endpoints Summary
8	W1-08b-agent-overview.png	Agent overview: inventory, MITRE ATT&CK, compliance, SCA
9	W1-09a-kali-test-events.png	Test events generated on Kali (su, sudo)
10	W1-09-threat-hunting-events.png	223 events in Threat Hunting
11 / 12	W1-10 / W1-10b alert JSON	Raw log and normalized fields (field mapping)
13	W1-12-indices.png	17 indices and sizes
14 / 15 / 16	W1-12c1 / W1-12c2 / W1-12d	Policy before, apply dialog, and after (Managed by policy = Yes)